

# CS 2601 Linear and Convex Optimization

## 15. Lagrange duality in general problems

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# Outline

- Dual function and dual problem
- Weak and strong duality
- Slater's condition
- KKT conditions revisited

## Lagrange dual function

Consider the general optimization problem (not necessarily convex),

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & h_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, k \\ & g_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, m \end{aligned} \tag{P}$$

The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{i=1}^k \lambda_i h_i(\mathbf{x}) + \sum_{j=1}^m \mu_j g_j(\mathbf{x})$$

The (Lagrange) dual function is

$$\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x} \in D} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x} \in D} \left( f(\mathbf{x}) + \sum_{i=1}^k \lambda_i h_i(\mathbf{x}) + \sum_{j=1}^m \mu_j g_j(\mathbf{x}) \right)$$

where  $D = \text{dom} f \cap (\bigcap_{i=1}^k \text{dom} h_i) \cap (\bigcap_{j=1}^m \text{dom} g_j)$  is the domain of the problem. We will downplay the role of  $D$  and focus on the case  $D = \mathbb{R}^n$ .

## Example

Given  $\mathbf{A} \in \mathbb{R}^{k \times n}$ ,

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) = \|\mathbf{x}\|^2 = \mathbf{x}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \mathbf{x}^T \mathbf{x} + \boldsymbol{\lambda}^T (\mathbf{A}\mathbf{x} - \mathbf{b}) - \boldsymbol{\mu}^T \mathbf{x}$$

Since  $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$  is convex in  $\mathbf{x}$ , its minimum satisfies

$$\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = 2\mathbf{x} + \mathbf{A}^T \boldsymbol{\lambda} - \boldsymbol{\mu} = \mathbf{0} \implies \mathbf{x} = \frac{1}{2}(\boldsymbol{\mu} - \mathbf{A}^T \boldsymbol{\lambda})$$

The dual function is

$$\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \mathcal{L}\left(\frac{1}{2}(\boldsymbol{\mu} - \mathbf{A}^T \boldsymbol{\lambda}), \boldsymbol{\lambda}, \boldsymbol{\mu}\right) = -\frac{1}{4}\|\boldsymbol{\mu} - \mathbf{A}^T \boldsymbol{\lambda}\|^2 - \mathbf{b}^T \boldsymbol{\lambda}$$

## Example

Given  $A \in \mathbb{R}^{k \times n}$ ,

$$\begin{array}{ll} \min_x & f(\mathbf{x}) = \mathbf{x}^T \mathbf{x} \\ \text{s.t.} & A\mathbf{x} = \mathbf{b} \end{array}$$

The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \mathbf{x}^T \mathbf{x} + \boldsymbol{\lambda}^T (A\mathbf{x} - \mathbf{b})$$

Since  $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$  is convex in  $\mathbf{x}$ , its minimum satisfies

$$\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = 2\mathbf{x} + A^T \boldsymbol{\lambda} = \mathbf{0} \implies \mathbf{x} = -\frac{1}{2} A^T \boldsymbol{\lambda}$$

The dual function is

$$\phi(\boldsymbol{\lambda}) = \mathcal{L}\left(-\frac{1}{2} A^T \boldsymbol{\lambda}, \boldsymbol{\lambda}\right) = -\frac{1}{4} \boldsymbol{\lambda}^T A A^T \boldsymbol{\lambda} - \mathbf{b}^T \boldsymbol{\lambda} = -\frac{1}{4} \|A^T \boldsymbol{\lambda}\|^2 - \mathbf{b}^T \boldsymbol{\lambda}$$

## Lower bound for optimal value

For any  $\lambda$  and any  $\mu \geq \mathbf{0}$ , the optimal value  $f^*$  of (P) is bounded by

$$f^* \geq \phi(\lambda, \mu)$$

**Proof.** Let  $X = \{\mathbf{x} : h_i(\mathbf{x}) = 0, \forall i; g_j(\mathbf{x}) \leq 0, \forall j\}$  be the feasible set.

- If  $X = \emptyset$ , then  $f^* = +\infty$ , trivially true.
- If  $X \neq \emptyset$ , for  $\mu \geq \mathbf{0}$  and  $\mathbf{x} \in X$ ,

$$f(\mathbf{x}) \geq f(\mathbf{x}) + \sum_{i=1}^k \lambda_i \underbrace{h_i(\mathbf{x})}_{=0} + \sum_{j=1}^m \underbrace{\mu_j g_j(\mathbf{x})}_{\leq 0} = \mathcal{L}(\mathbf{x}, \lambda, \mu)$$

Minimizing over  $\mathbf{x}$ ,

$$f^* = \inf_{\mathbf{x} \in X} f(\mathbf{x}) \geq \inf_{\mathbf{x} \in X} \mathcal{L}(\mathbf{x}, \lambda, \mu) \geq \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda, \mu) = \phi(\lambda, \mu)$$

## Concavity of dual function

The dual function is always concave, whether the primal problem (P) is convex or not.

**Proof.** Note  $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$  is affine in  $(\boldsymbol{\lambda}, \boldsymbol{\mu})$ . Thus  $\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$  is the pointwise infimum of a family of affine functions indexed by  $\mathbf{x}$ , and hence concave. (Recall the pointwise supremum of convex functions is convex).

$$\begin{aligned}\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) &= \inf_{\mathbf{x} \in D} \left( f(\mathbf{x}) + \sum_{i=1}^k \lambda_i h_i(\mathbf{x}) + \sum_{j=1}^m \mu_j g_j(\mathbf{x}) \right) \\ &= - \underbrace{\sup_{\mathbf{x} \in D} \left( -f(\mathbf{x}) - \sum_{i=1}^k \lambda_i h_i(\mathbf{x}) - \sum_{j=1}^m \mu_j g_j(\mathbf{x}) \right)}_{\text{pointwise supremum of convex (affine) functions in } (\boldsymbol{\lambda}, \boldsymbol{\mu})}\end{aligned}$$

**Example.**  $\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = -\frac{1}{4} \|\boldsymbol{\mu} - \mathbf{A}^T \boldsymbol{\lambda}\|^2 - \mathbf{b}^T \boldsymbol{\lambda}$  is concave.

# Lagrange dual problem

To find the best lower bound given by the dual function

$$f^* \geq \phi(\boldsymbol{\lambda}, \boldsymbol{\mu})$$

solve the (Lagrange) dual problem associated with the primal problem (P),

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) \\ \text{s.t.} \quad & \boldsymbol{\mu} \geq \mathbf{0} \end{aligned} \tag{D}$$

The dual problem (D) is **always** convex, whether or not (P) is convex.

$(\boldsymbol{\lambda}, \boldsymbol{\mu})$  is **dual feasible** if  $\boldsymbol{\mu} \geq \mathbf{0}$  and  $\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) > -\infty$ .

**Note.** The domain of a convex function  $f$  is  $\text{dom } f = \{\mathbf{x} : f(\mathbf{x}) < +\infty\}$ , while the domain of a concave function  $f$  is  $\text{dom } f = \{\mathbf{x} : f(\mathbf{x}) > -\infty\}$ . Thus the condition  $\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) > -\infty$  just means  $(\boldsymbol{\lambda}, \boldsymbol{\mu}) \in \text{dom } \phi$ .



## Example

Recall the dual problem of the following LP

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{Gx} \geq \mathbf{h}\end{array}$$

is

$$\begin{array}{ll}\max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} & \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \begin{cases} \boldsymbol{\lambda}^T \mathbf{b} + \boldsymbol{\mu}^T \mathbf{h}, & \text{if } \mathbf{A}^T \boldsymbol{\lambda} + \mathbf{G}^T \boldsymbol{\mu} = \mathbf{c} \\ -\infty, & \text{otherwise} \end{cases} \\ \text{s.t.} & \boldsymbol{\mu} \geq \mathbf{0}\end{array}$$

$(\boldsymbol{\lambda}, \boldsymbol{\mu})$  is dual feasible if  $\boldsymbol{\mu} \geq \mathbf{0}$  and  $\mathbf{A}^T \boldsymbol{\lambda} + \mathbf{G}^T \boldsymbol{\mu} = \mathbf{c}$ , which just means it is feasible for the dual LP,

$$\begin{array}{ll}\max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} & \psi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \boldsymbol{\lambda}^T \mathbf{b} + \boldsymbol{\mu}^T \mathbf{h} \\ \text{s.t.} & \mathbf{A}^T \boldsymbol{\lambda} + \mathbf{G}^T \boldsymbol{\mu} = \mathbf{c} \\ & \boldsymbol{\mu} \geq \mathbf{0}\end{array}$$

## Example

Recall the dual problem of the following problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) = \mathbf{x}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

is

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = -\frac{1}{4} \|\boldsymbol{\mu} - \mathbf{A}^T \boldsymbol{\lambda}\|^2 - \mathbf{b}^T \boldsymbol{\lambda} \\ \text{s.t.} \quad & \boldsymbol{\mu} \geq \mathbf{0} \end{aligned}$$

$(\boldsymbol{\lambda}, \boldsymbol{\mu})$  is dual feasible if  $\boldsymbol{\mu} \geq \mathbf{0}$ , as there is no implicit constraint in  $\phi$ .

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# Weak and strong duality

Denote by  $f^*$  and  $\phi^*$  the primal and dual optimal values, i.e.

$$f^* = \inf_{\mathbf{x} \in X} f(\mathbf{x}), \quad \phi^* = \sup_{\boldsymbol{\lambda}, \boldsymbol{\mu}: \boldsymbol{\mu} \geq \mathbf{0}} \phi(\boldsymbol{\lambda}, \boldsymbol{\mu})$$

Weak duality:  $f^* \geq \phi^*$

- **always** holds.

**Proof.** Recall  $f^* \geq \phi(\boldsymbol{\lambda}, \boldsymbol{\mu})$  for any  $\boldsymbol{\lambda}$  and any  $\boldsymbol{\mu} \geq \mathbf{0}$ . Weak duality follows by maximizing over  $\boldsymbol{\lambda}$  and  $\boldsymbol{\mu} \geq \mathbf{0}$ .

- $f^* - \phi^*$  is called the (optimal) duality gap of the problem.

Strong duality:  $f^* = \phi^*$

- does **not** hold in general.
- typically holds for convex problems under various conditions known as **constraint qualifications**, e.g. Slater's condition.
- may also hold for nonconvex problems.
- can solve the dual problem instead if it is easier than the primal.

## Duality gap

Given primal feasible  $x$  and dual feasible  $(\lambda, \mu)$ , the difference

$$f(x) - \phi(\lambda, \mu)$$

is called the **duality gap** associated with  $x$  and  $(\lambda, \mu)$ .

Note

$$\phi(\lambda, \mu) \leq \phi^* \leq f^* \leq f(x)$$

If the duality gap is zero, i.e.  $f(x) = \phi(\lambda, \mu)$ , then all inequalities become equalities, so  $x$  is primal optimal, and  $(\lambda, \mu)$  is dual optimal.

If the gap  $f(x) - \phi(\lambda, \mu) \leq \epsilon$ , then the dual solution  $(\lambda, \mu)$  serves as a **proof** or **certificate** that  $x$  is  $\epsilon$ -suboptimal,

$$f(x) - f^* \leq f(x) - \phi(\lambda, \mu) \leq \epsilon$$

When strong duality holds, this can serve as a stopping criterion in an iterative algorithm, i.e. stop when  $f(x) - \phi(\lambda, \mu) \leq \epsilon$  for some  $(\lambda, \mu)$ .

## Example

$$\begin{aligned} \min_{x \in \mathbb{R}} \quad & f(x) = x^2 \\ \text{s.t.} \quad & x \leq a \end{aligned}$$

The dual function is

$$\phi(\mu) = \inf_x [x^2 + \mu(x - a)] = -\frac{\mu^2}{4} - a\mu$$

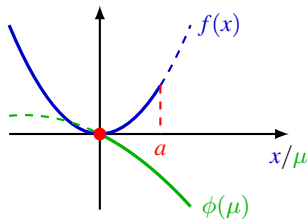
The dual problem is

$$\begin{aligned} \max_{\mu \in \mathbb{R}} \quad & \phi(\mu) = -\frac{\mu^2}{4} - a\mu \\ \text{s.t.} \quad & \mu \geq 0 \end{aligned}$$

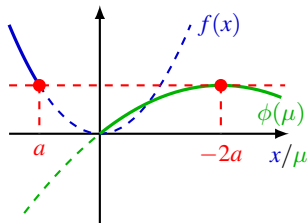
The primal and dual optimal values are

1. If  $a \geq 0$ ,  $f^* = f(0) = \phi^* = \phi(0) = 0$
2. If  $a \leq 0$ ,  $f^* = f(a) = \phi^* = \phi(-2a) = a^2$

Strong duality holds in both cases.



Case 1.  $a \geq 0$



Case 2.  $a \leq 0$

## Example

Consider

$$\begin{array}{ll}\min_{x \in \mathbb{R}} & f(x) = x^3 \\ \text{s.t.} & x \geq 0\end{array}$$

The optimal value is  $f^* = f(0) = 0$ .

The dual function is

$$\phi(\mu) = \inf_x [x^3 - \mu x] = -\infty$$

so the dual optimal value is

$$\phi^* = \sup_{\mu \geq 0} \phi(\mu) = -\infty$$

The duality gap is infinite. In particular, strong duality does not hold.

## Example

Consider

$$\begin{aligned} \min_{x \in \mathbb{R}} \quad & f(x) = \begin{cases} -x^2 - x + \frac{3}{4}, & |x| \leq \frac{1}{2} \\ x^2 - x + \frac{1}{4}, & |x| \geq \frac{1}{2} \end{cases} \\ \text{s.t.} \quad & x \leq 0 \end{aligned}$$

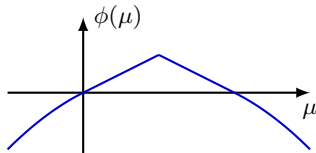
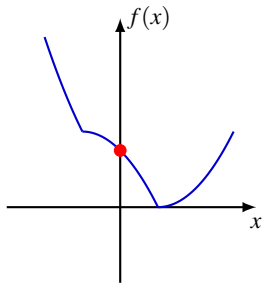
The primal optimal value is  $f^* = f(0) = \frac{3}{4}$ .

The dual function is

$$\phi(\mu) = \inf_x [f(x) + \mu x] = \begin{cases} \frac{1 - |\mu - 1|}{2}, & |\mu - 1| \leq 1 \\ \frac{1 - (\mu - 1)^2}{4}, & |\mu - 1| \geq 1 \end{cases}$$

The dual optimal value is  $\phi^* = \phi(1) = \frac{1}{2}$ .

The duality gap is  $f^* - \phi^* = \frac{1}{4}$ .





## Example (cont'd)

To compute the dual function, note

$$\mathcal{L}(x, \mu) = f(x) + \mu x = \begin{cases} -x^2 + (\mu - 1)x + \frac{3}{4}, & |x| \leq \frac{1}{2} \\ x^2 + (\mu - 1)x + \frac{1}{4}, & |x| > \frac{1}{2} \end{cases}$$

Since  $y = -x^2 + (\mu - 1)x + \frac{3}{4}$  is a parabola opening down,

$$\phi_1(\mu) = \inf_{|x| \leq \frac{1}{2}} \mathcal{L}(x, \mu) = \min \left\{ \mathcal{L}\left(\frac{1}{2}, \mu\right), \mathcal{L}\left(-\frac{1}{2}, \mu\right) \right\} = \frac{1 - |\mu - 1|}{2}$$

Since  $y = x^2 + (\mu - 1)x + \frac{1}{4}$  is a parabola opening up,

$$\phi_2(\mu) = \inf_{|x| \geq \frac{1}{2}} \mathcal{L}(x, \mu) = \begin{cases} \mathcal{L}\left(\frac{1-\mu}{2}, \mu\right) = \frac{1-(\mu-1)^2}{4}, & |\mu - 1| \geq 1 \\ \min \left\{ \mathcal{L}\left(\frac{1}{2}, \mu\right), \mathcal{L}\left(-\frac{1}{2}, \mu\right) \right\} = \frac{1-|\mu-1|}{2}, & |\mu - 1| \leq 1 \end{cases}$$

Thus

$$\phi(\mu) = \min\{\phi_1(\mu), \phi_2(\mu)\} = \phi_2(\mu)$$

## Example (cont'd)

By definition of dual function,

$$\phi(\mu) = \inf_x [f(x) + \mu x] \leq f(x) + \mu x$$

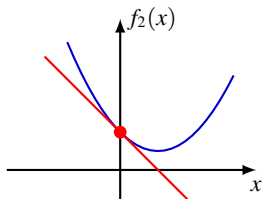
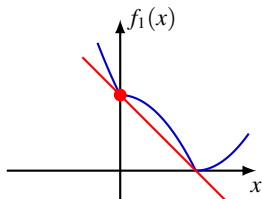
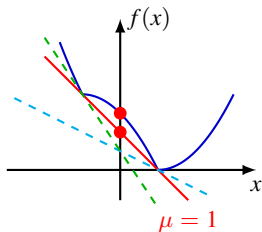
Rearranging,

$$\ell(x) \triangleq -\mu x + \phi(\mu) \leq f(x)$$

Note  $\ell(x)$  is a line with slope  $-\mu$  and intercept  $\phi(\mu)$  that supports  $\text{epi} f$ .

The dual optimal value  $\phi^*$  is the largest intercept of such lines. We can see pictorially there is a gap.

This also give us intuition about why strong duality may hold for nonconvex problem, and why it usually holds for convex problems.



## Example

$$\min_{x \in \mathbb{R}} f(x) = \begin{cases} -\sqrt{x}, & x > 0 \\ 1 & x = 0 \\ +\infty, & x < 0 \end{cases}$$

$$\text{s.t. } x \leq 0$$

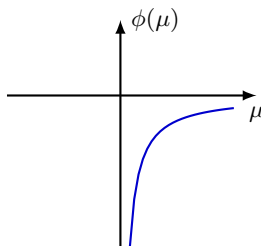
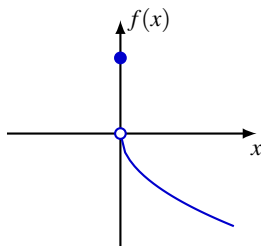
The primal optimal value is  $f^* = f(0) = 1$ .

The dual function is

$$\phi(\mu) = \inf_x [f(x) + \mu x] = \begin{cases} -\frac{1}{4\mu}, & \mu > 0 \\ -\infty, & \mu \leq 0 \end{cases}$$

The dual optimal value is  $\phi^* = 0$ , which is not attainable.

This is a convex problem with nonzero duality gap  $f^* - \phi^* = 1$ , a nontypical case.



## Example

$$\min_{x \in \mathbb{R}} f(x) = \begin{cases} -\sqrt{x}, & x > 0 \\ 1 & x = 0 \\ +\infty, & x < 0 \end{cases}$$

$$\text{s.t. } x \leq a$$

where  $a > 0$ .

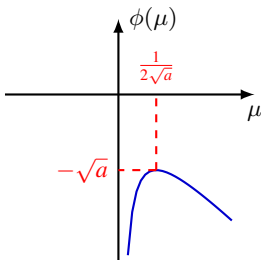
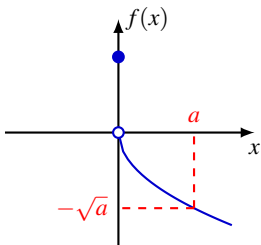
The primal optimal value is  $f^* = f(a) = -\sqrt{a}$ .

The dual function is

$$\phi(\mu) = \inf_x [f(x) + \mu(x-a)] = \begin{cases} -\frac{1}{4\mu} - a\mu, & \mu > 0 \\ -\infty, & \mu \leq 0 \end{cases}$$

The dual optimal value is  $\phi^* = \phi(\frac{1}{2\sqrt{a}}) = -\sqrt{a}$

Strong duality holds in this case.



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# Slater's condition for convex problems

Consider a convex problem,

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & g_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, m \\ & \mathbf{h}(\mathbf{x}) = \mathbf{Ax} - \mathbf{b} = \mathbf{0}\end{array} \quad (\text{CP})$$

with domain  $D = \text{dom} f \cap (\bigcap_{j=1}^m \text{dom} g_j)$ .

**Slater's condition.** The above problem is **strictly feasible**, i.e.

$$\exists \mathbf{x} \in \text{int } D^1 \quad \text{s.t.} \quad g_j(\mathbf{x}) < 0 \text{ for } j = 1, 2, \dots, m, \quad \mathbf{Ax} = \mathbf{b}$$

**Refined Slater's condition.** If some  $g_j$  are affine, the requirement  $g_j(\mathbf{x}) < 0$  can be relaxed to feasibility  $g_j(\mathbf{x}) \leq 0$  for those  $g_j$ .

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<sup>1</sup> $\text{int } D$  stands for the interior of  $D$ .  $\mathbf{x} \in \text{int } D$  if there exists  $\delta > 0$  s.t.  $B(\mathbf{x}, \delta) \subset D$ .  
Again we focus on the case  $D = \mathbb{R}^n$ , so the requirement  $\mathbf{x} \in \text{int } D$  is always satisfied. <sup>21</sup>

# Slater's Theorem

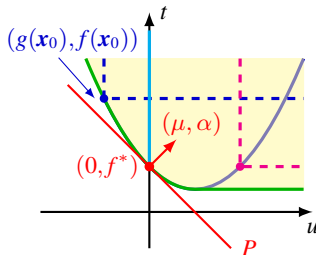
**Slater's Theorem.** Strong duality holds for (CP) under (refined) Slater's condition. Furthermore, if  $\phi^* > -\infty$ , it is attained by some  $(\lambda^*, \mu^*)$ .

**Proof idea.** Consider the case with only one inequality constraint  $g$ . Let

$$C = \{(u, t) : \exists \mathbf{x} \in D \text{ s.t. } g(\mathbf{x}) \leq u, f(\mathbf{x}) \leq t\}$$

$C$  is convex and  $(0, f(\mathbf{x})) \in C$  for feasible  $\mathbf{x}$ .  $C$  has a supporting hyperplane  $P$  at  $(0, f^*) \in \partial C$ ,

$$\mu u + \alpha t \geq \alpha f^*, \quad \forall (u, t) \in C$$



Letting  $u, t \rightarrow +\infty$  shows  $\mu \geq 0, \alpha \geq 0$ . By Slater's condition,  $\exists \mathbf{x}_0 \in D$  s.t.  $g(\mathbf{x}_0) < 0$ , so  $P$  is non-vertical, i.e.  $\alpha \neq 0$ . Since  $(g(\mathbf{x}), f(\mathbf{x})) \in C$ ,

$$f^* \leq \mu^* g(\mathbf{x}) + f(\mathbf{x}) = \mathcal{L}(\mathbf{x}, \mu^*), \text{ where } \mu^* = \frac{\mu}{\alpha} \geq 0$$

Minimizing over  $\mathbf{x} \in D$ ,  $f^* \leq \phi(\mu^*) \leq \phi^*$ . Weak duality then implies  $f^* = \phi(\mu^*) = \phi^*$ . The condition  $\mathbf{x}_0 \in \text{int } D$  will be used to deal with  $\mathbf{h}$ .

## Proof

If  $f^* = -\infty$ , then  $\phi^* = f^* = -\infty$  by weak duality. Assume  $f^* > -\infty$ . Since (CP) is strictly feasible by Slater's condition,  $f^* < +\infty$ . Also assume  $A \in \mathbb{R}^{k \times n}$  has  $\text{rank } A = k$ , by removing redundant constraints. Now let

$$C = \{(u, v, t) : \exists x \in D \text{ s.t. } g(x) \leq u, h(x) = v, f(x) \leq t\}$$

1.  $C$  is convex. The proof is similar to that on slide 7 of §4 part 2.
2.  $C \neq \emptyset$  and  $(0, 0, f^*) \in \partial C$ .
  - 2.1 Note  $(0, 0, t) \in C$  iff  $\exists x \in D$  s.t.  $g(x) \leq 0, h(x) = 0, f(x) \leq t$ , i.e. iff  $\exists x \in X$  s.t.  $f(x) \leq t$ . In particular,  $(0, 0, f(x)) \in C$  for  $x \in X$ .
  - 2.2 Since  $f^* = \inf_{x \in X} f(x)$ , there exists a sequence  $\{x_i\} \subset X$  s.t.  $f(x_i) \rightarrow f^*$ . Since  $C \ni (0, 0, f(x_i)) \rightarrow (0, 0, f^*)$ , we have  $(0, 0, f^*) \in \overline{C}$ .
  - 2.3  $(0, 0, t) \notin C$  for any  $t < f^*$ . Thus  $(0, 0, f^*) \notin \text{int } C$  and  $(0, 0, f^*) \in \partial C$ .
3. There exists a supporting hyperplane at  $(0, 0, f^*) \in \partial C$ , i.e. there exists  $(\mu, \lambda, \alpha) \neq 0$  s.t. for all  $(u, v, t) \in C$ ,

$$(\mu, \lambda, \alpha) \cdot (u, v, t) = \mu^T u + \lambda^T v + \alpha t \geq (\mu, \lambda, \alpha) \cdot (0, 0, f^*) = \alpha f^*$$



## Proof (cont'd)

4. Since  $\mathbf{u}, t$  can be arbitrarily large for  $(\mathbf{u}, \mathbf{v}, t) \in C$ , letting  $\mathbf{u}, t \rightarrow \infty$  yields  $\boldsymbol{\mu} \geq \mathbf{0}, \alpha \geq 0$ .
5. Since  $(\mathbf{g}(\mathbf{x}), \mathbf{h}(\mathbf{x}), f(\mathbf{x})) \in C$ ,

$$\boldsymbol{\mu}^T \mathbf{g}(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{h}(\mathbf{x}) + \alpha f(\mathbf{x}) \geq \alpha f^*, \quad \forall \mathbf{x} \in D$$

6. If  $\alpha \neq 0$ , then

$$f^* \leq f(\mathbf{x}) + (\boldsymbol{\lambda}^*)^T \mathbf{h}(\mathbf{x}) + (\boldsymbol{\mu}^*)^T \mathbf{g}(\mathbf{x}) = \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$$

where  $\boldsymbol{\mu}^* = \boldsymbol{\mu}/\alpha \geq \mathbf{0}$  and  $\boldsymbol{\lambda}^* = \boldsymbol{\lambda}/\alpha$ . Minimizing over  $\mathbf{x}$ ,

$$f^* \leq \phi(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \leq \phi^*$$

Weak duality then implies

$$f^* = \phi(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \phi^*$$

i.e. strong duality holds, and the dual optimal is attained.

## Proof (cont'd)

7. Now we show Slater's condition implies  $\alpha \neq 0$ . Suppose  $\alpha = 0$ .

7.1 By 5,

$$\mu^T g(x) + \lambda^T h(x) \geq 0, \quad \forall x \in D$$

7.2 Let  $x_0$  satisfy Slater's condition, i.e.  $x_0 \in \text{int } D$ ,  $h(x_0) = \mathbf{0}$ , and  $g(x_0) < \mathbf{0}$ . Since  $\mu \geq \mathbf{0}$ ,

$$\underbrace{\mu^T g(x_0)}_{\leq 0} + \lambda^T \underbrace{h(x_0)}_{=0} \geq 0 \implies \mu = \mathbf{0}$$

7.3 By 7.1 and 7.2,

$$\lambda^T h(x) \geq 0, \quad \forall x \in D$$

7.4 Since  $x_0 \in \text{int } D$ , there exists  $\delta > 0$  s.t.  $x_0 + z \in D$  for all  $z \in B(\mathbf{0}, \delta)$ . Recalling  $h(x) = Ax - b$  and  $h(x_0) = \mathbf{0}$ , we have

$$\lambda^T A z = \lambda^T [h(x_0 + z) - h(x_0)] \geq 0, \quad \forall z \in B(\mathbf{0}, \delta) \implies A^T \lambda = \mathbf{0}$$

7.5 Since  $A$  has full row rank,  $\lambda = \mathbf{0}$ . Thus  $(\mu, \lambda, \alpha) = \mathbf{0}$ , contradicting  $(\mu, \lambda, \alpha) \neq \mathbf{0}$  given by 3.

## Example

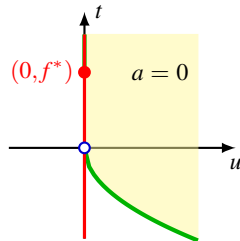
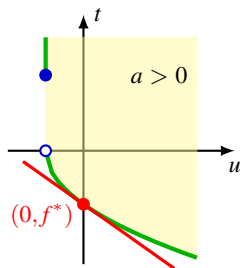
$$\min_{x \in \mathbb{R}} f(x) = \begin{cases} -\sqrt{x}, & x > 0 \\ 1 & x = 0 \\ +\infty, & x < 0 \end{cases}$$

$$\text{s.t. } x \leq a$$

is a convex problem with domain  $D = [0, \infty)$ .

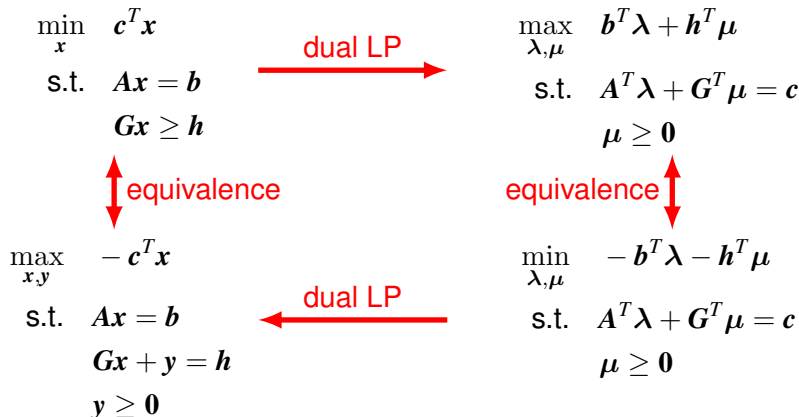
Note  $\text{int } D = (0, \infty)$ .

- If  $a > 0$ , Slater's condition is satisfied, e.g.  $\frac{a}{2} \in \text{int } D$  and  $\frac{a}{2} < a$ , so strong duality must hold.
- If  $a = 0$ , no point in  $\text{int } D$  is feasible. Slater's Theorem is not applicable<sup>2</sup>, and it turns out that strong duality does not hold.



<sup>2</sup>Slater's condition is only a sufficient condition for strong duality. It is not necessary.

## Example: Strong duality for LP



- Essentially, dual of dual is primal.
- By refined Slater's condition, strong duality holds if either the primal or the dual is feasible.
- When either  $f^*$  or  $\phi^*$  is finite, then  $f^* = \phi^*$  and they are both attained.

## Example: Strong duality for LP (cont'd)

There are four possibilities

1. Primal feasible, dual feasible,  $-\infty < \phi^* = f^* < +\infty$
2. Primal feasible, dual infeasible,  $f^* = \phi^* = -\infty$

$$\begin{array}{ll}\min & x_1 - 2x_2 \\ \text{s.t.} & x_1 - x_2 = -1 \\ & x_1, x_2 \geq 0\end{array}\qquad \begin{array}{ll}\max & \lambda \\ \text{s.t.} & \lambda + \mu_1 = 1 \\ & -\lambda + \mu_2 = -2 \\ & \mu_1, \mu_2 \geq 0\end{array}$$

3. Primal infeasible, dual feasible,  $f^* = \phi^* = +\infty$
4. Primal infeasible, dual infeasible,  $f^* = +\infty, \phi^* = -\infty$

$$\begin{array}{ll}\min & x_1 - 2x_2 \\ \text{s.t.} & x_1 - x_2 \leq 1 \\ & -x_1 + x_2 \leq -2\end{array}\qquad \begin{array}{ll}\max & -\mu_1 + 2\mu_2 \\ \text{s.t.} & -\mu_1 + \mu_2 = 1 \\ & \mu_1 - \mu_2 = -2 \\ & \mu_1, \mu_2 \geq 0\end{array}$$

**Note.** No duality gap in Case 2 and Case 3, but  $f^* - \phi^*$  is undefined.

## Example: Dual formulation of SVM

Recall the primal formulation of SVM,

$$\begin{aligned} \min_{\mathbf{w}, b, \boldsymbol{\xi}} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 + C \mathbf{1}^T \boldsymbol{\xi} \\ \text{s. t.} \quad & y_i(\mathbf{x}_i^T \mathbf{w} + b) \geq 1 - \xi_i, \quad i = 1, 2, \dots, n \\ & \boldsymbol{\xi} \geq \mathbf{0} \end{aligned}$$

where  $C > 0$  is a hyperparameter, and  $\mathbf{1}$  is the vector of all 1's.

- convex problem with affine constraints.
- always feasible. Indeed, given any  $\mathbf{w}, b$ ,

$$\xi_i = [1 - y_i(\mathbf{w}^T \mathbf{x}_i + b)]^+, \quad i = 1, 2, \dots, n$$

yields a feasible solution  $(\mathbf{w}, b, \boldsymbol{\xi})$ , where  $(x)^+ = \max\{x, 0\}$ .

- strong duality holds by refined Slater's condition
- can solve the dual problem instead, which turns out to be useful!

## Example: Dual formulation of SVM (cont'd)

The Lagrangian is

$$\begin{aligned}\mathcal{L}(\mathbf{w}, b, \boldsymbol{\xi}, \boldsymbol{\mu}, \boldsymbol{\alpha}) &= \frac{1}{2} \|\mathbf{w}\|_2^2 + C \mathbf{1}^T \boldsymbol{\xi} + \sum_{i=1}^n \mu_i [1 - \xi_i - y_i (\mathbf{x}_i^T \mathbf{w} + b)] - \boldsymbol{\alpha}^T \boldsymbol{\xi} \\ &= \frac{1}{2} \|\mathbf{w}\|_2^2 - \left( \sum_{i=1}^n y_i \mu_i \mathbf{x}_i \right)^T \mathbf{w} - \boldsymbol{\mu}^T \mathbf{y} b + (C \mathbf{1} - \boldsymbol{\mu} - \boldsymbol{\alpha})^T \boldsymbol{\xi} + \mathbf{1}^T \boldsymbol{\mu}\end{aligned}$$

Minimizing over  $\mathbf{w}, b, \boldsymbol{\xi}$  yields the dual function ( $\mathbf{w} = \sum_{i=1}^n y_i \mu_i \mathbf{x}_i$ ),

$$\phi(\boldsymbol{\mu}, \boldsymbol{\alpha}) = \begin{cases} \mathbf{1}^T \boldsymbol{\mu} - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \mu_i \mu_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j, & \text{if } \boldsymbol{\mu}^T \mathbf{y} = 0, C \mathbf{1} - \boldsymbol{\mu} - \boldsymbol{\alpha} = \mathbf{0} \\ -\infty, & \text{otherwise} \end{cases}$$

The dual problem is

$$\begin{aligned}\max_{\boldsymbol{\mu}, \boldsymbol{\alpha}} \quad & \phi(\boldsymbol{\mu}, \boldsymbol{\alpha}) \\ \text{s. t.} \quad & \boldsymbol{\mu} \geq \mathbf{0}, \boldsymbol{\alpha} \geq \mathbf{0}\end{aligned}$$

## Example: Dual formulation of SVM (cont'd)

Making the constraints explicit, we obtain the equivalent problem,

$$\begin{aligned} \max_{\boldsymbol{\mu}, \boldsymbol{\alpha}} \quad & \mathbf{1}^T \boldsymbol{\mu} - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \mu_i \mu_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j \\ \text{s. t.} \quad & \boldsymbol{\mu}^T \mathbf{y} = 0 \\ & \boldsymbol{\mu} + \boldsymbol{\alpha} = C \mathbf{1} \\ & \boldsymbol{\mu} \geq \mathbf{0}, \boldsymbol{\alpha} \geq \mathbf{0} \end{aligned}$$

Eliminating  $\boldsymbol{\alpha}$ , we obtain the following dual formulation of SVM,

$$\begin{aligned} \max_{\boldsymbol{\mu}} \quad & \mathbf{1}^T \boldsymbol{\mu} - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \mu_i \mu_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j \\ \text{s. t.} \quad & \boldsymbol{\mu}^T \mathbf{y} = 0 \\ & \mathbf{0} \leq \boldsymbol{\mu} \leq C \mathbf{1} \end{aligned}$$

Can be solved efficiently by an algorithm called **Sequential Minimal Optimization (SMO)**. Also amenable to further generalization using the **kernel trick** that replaces  $\mathbf{x}_i^T \mathbf{x}_j$  by a **kernel (function)**  $K(\mathbf{x}_i, \mathbf{x}_j)$ .



# Outline

- Dual function and dual problem
- Weak and strong duality
- Slater's condition
- KKT conditions revisited

# KKT conditions for convex problems

Consider a differentiable convex problem and its dual,

$$\begin{array}{ll} \min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ & \mathbf{h}(\mathbf{x}) = \mathbf{Ax} - \mathbf{b} = \mathbf{0} \end{array} \quad \left| \quad \begin{array}{ll} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} & \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \\ \text{s.t.} & \boldsymbol{\mu} \geq \mathbf{0} \end{array} \right.$$

**Theorem.** KKT conditions hold at  $\mathbf{x}^*$  with Lagrange multipliers  $\boldsymbol{\lambda}^*$ ,  $\boldsymbol{\mu}^*$ ,

1. (primal feasibility)  $\mathbf{h}(\mathbf{x}^*) = \mathbf{0}, \mathbf{g}(\mathbf{x}^*) \leq \mathbf{0}$
2. (dual feasibility)  $\boldsymbol{\mu}^* \geq \mathbf{0}$
3. (stationarity)  $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \mathbf{0}$
4. (complementary slackness)  $\mu_j^* g_j(\mathbf{x}^*) = 0, j = 1, 2, \dots, m$

if and only if all the following conditions hold,

1. strong duality holds, i.e.  $f^* = \phi^*$
2.  $\mathbf{x}^*$  is a primal optimal solution, i.e.  $f^* = f(\mathbf{x}^*)$
3.  $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$  is a dual optimal solution, i.e.  $\phi^* = \phi(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$

## Proof of necessity

Assume KKT holds at  $\mathbf{x}^*$  with Lagrange multipliers  $\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*$ .

- Since  $\boldsymbol{\mu}^* \geq \mathbf{0}$ ,  $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = f(\mathbf{x}) + (\boldsymbol{\lambda}^*)^T \mathbf{h}(\mathbf{x}) + (\boldsymbol{\mu}^*)^T \mathbf{g}(\mathbf{x})$  is convex in  $\mathbf{x}$ .
- The stationarity condition  $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \mathbf{0}$  implies  $\mathbf{x}^*$  is a global minimum of  $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ , i.e.

$$\mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \phi(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$$

- By primal feasibility and complementary slackness,

$$\mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = f(\mathbf{x}^*) + (\boldsymbol{\lambda}^*)^T \underbrace{\mathbf{h}(\mathbf{x}^*)}_{=\mathbf{0}} + \underbrace{(\boldsymbol{\mu}^*)^T \mathbf{g}(\mathbf{x}^*)}_{=\mathbf{0}} = f(\mathbf{x}^*)$$

so

$$f(\mathbf{x}^*) = \phi(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$$

- By the discussion on slide 12,  $\mathbf{x}^*$  is primal optimal,  $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$  is dual optimal and strong duality holds.

## Proof of sufficiency

Assume strong duality holds,  $\mathbf{x}^*$  is primal optimal, and  $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$  is dual optimal. We only need to show the stationarity condition and the complementary slackness condition.

$$\begin{aligned} f^* &= \phi^* && \text{(strong duality)} \\ &= \phi(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) && \text{(dual optimality of } (\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \text{)} \\ &= \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) && \text{(definition of dual function)} \\ &\leq \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) && \text{(definition of infimum)} \\ &= f(\mathbf{x}^*) + \underbrace{(\boldsymbol{\lambda}^*)^T \mathbf{h}(\mathbf{x}^*)}_{=0} + \underbrace{(\boldsymbol{\mu}^*)^T \mathbf{g}(\mathbf{x}^*)}_{\geq 0} \underbrace{\mathbf{g}(\mathbf{x}^*)}_{\leq 0} \\ &\leq f(\mathbf{x}^*) && \text{(primal and dual feasibility of } \mathbf{x}^*, \boldsymbol{\mu}^* \text{)} \\ &= f^* && \text{(primal optimality of } \mathbf{x}^* \text{)} \end{aligned}$$

So both inequality holds with equality. The first implies  $\mathbf{x}^*$  is a minimum of  $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ , so  $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \mathbf{0}$ . The second implies  $(\boldsymbol{\mu}^*)^T \mathbf{g}(\mathbf{x}^*) = \mathbf{0}$ , so  $\mu_j g_j(\mathbf{x}^*) = 0$  for  $j = 1, 2, \dots, m$ .

## KKT conditions under Slater's condition

**Corollary.** If  $\mathbf{x}^*$  is a minimum of a differentiable convex problem that satisfies (refined) Slater's condition, then the KKT conditions hold at  $\mathbf{x}^*$  with some Lagrange multipliers  $\lambda^*, \mu^*$ .

**Proof.** Under (refined) Slater's condition, strong duality holds. Since

$$\phi^* = f^* = f(\mathbf{x}^*) > -\infty$$

$\phi^*$  is attained at some  $\lambda^*, \mu^*$ . By the previous theorem, the KKT conditions hold at  $\mathbf{x}^*$  with  $\lambda^*, \mu^*$  being the Lagrange multipliers.